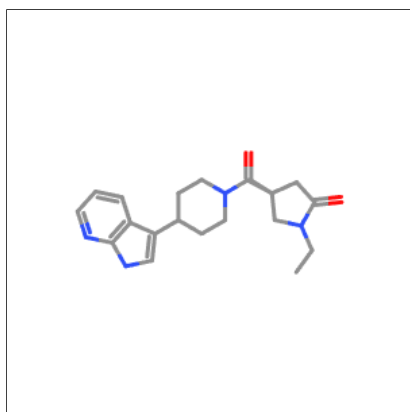
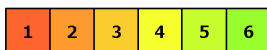


## Oral toxicity prediction results for input compound



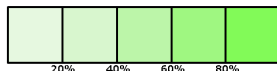
Predicted LD50: 200mg/kg

Predicted Toxicity Class: 3



Average similarity: 54.93%

Prediction accuracy: 67.38%



|                                           |                                        |
|-------------------------------------------|----------------------------------------|
| Name                                      | CCN1C[C@@H](C(=O)N2CCC(c3c[nH]c4)C2=O) |
| Molweight                                 | 340.42                                 |
| Number of hydrogen bond acceptors         | 5                                      |
| Number of hydrogen bond donors            | 1                                      |
| Number of atoms                           | 25                                     |
| Number of bonds                           | 28                                     |
| Number of rotatable bonds                 | 4                                      |
| Molecular refractivity                    | 103.73                                 |
| Topological Polar Surface Area            | 69.3                                   |
| octanol/water partition coefficient(logP) | 2.01                                   |

## Toxicity Model Report

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| Classification                             | Target                                                                                                | Shorthand     | Prediction | Probability |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------|------------|-------------|
| Organ toxicity                             | <a href="#">Hepatotoxicity</a>                                                                        | dili          | Inactive   | 0.83        |
| Organ toxicity                             | <a href="#">Neurotoxicity</a>                                                                         | neuro         | Active     | 0.91        |
| Organ toxicity                             | <a href="#">Nephrotoxicity</a>                                                                        | nephro        | Inactive   | 0.76        |
| Organ toxicity                             | <a href="#">Respiratory toxicity</a>                                                                  | respi         | Active     | 0.86        |
| Organ toxicity                             | <a href="#">Cardiotoxicity</a>                                                                        | cardio        | Inactive   | 0.91        |
| Toxicity end points                        | <a href="#">Carcinogenicity</a>                                                                       | carcino       | Inactive   | 0.61        |
| Toxicity end points                        | <a href="#">Immunotoxicity</a>                                                                        | immuno        | Active     | 0.98        |
| Toxicity end points                        | <a href="#">Mutagenicity</a>                                                                          | mutagen       | Inactive   | 0.57        |
| Toxicity end points                        | <a href="#">Cytotoxicity</a>                                                                          | cyto          | Inactive   | 0.69        |
| Toxicity end points                        | <a href="#">BBB-barrier</a>                                                                           | bbb           | Active     | 0.93        |
| Toxicity end points                        | <a href="#">Ecotoxicity</a>                                                                           | eco           | Inactive   | 0.51        |
| Toxicity end points                        | <a href="#">Clinical toxicity</a>                                                                     | clinical      | Active     | 0.77        |
| Toxicity end points                        | <a href="#">Nutritional toxicity</a>                                                                  | nutri         | Inactive   | 0.57        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Aryl hydrocarbon Receptor (AhR)</a>                                                       | nr_ahr        | Inactive   | 0.82        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Androgen Receptor (AR)</a>                                                                | nr_ar         | Inactive   | 0.96        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Androgen Receptor Ligand Binding Domain (AR-LBD)</a>                                      | nr_ar_lbd     | Inactive   | 0.98        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Aromatase</a>                                                                             | nr_aromatase  | Inactive   | 0.90        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Estrogen Receptor Alpha (ER)</a>                                                          | nr_er         | Inactive   | 0.94        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Estrogen Receptor Ligand Binding Domain (ER-LBD)</a>                                      | nr_er_lbd     | Inactive   | 0.98        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Peroxisome Proliferator Activated Receptor Gamma (PPAR-Gamma)</a>                         | nr_ppar_gamma | Inactive   | 0.91        |
| Tox21-Stress response pathways             | <a href="#">Nuclear factor (erythroid-derived 2)-like 2/antioxidant responsive element (nrf2/ARE)</a> | sr_are        | Inactive   | 0.98        |
| Tox21-Stress response pathways             | <a href="#">Heat shock factor response element (HSE)</a>                                              | sr_hse        | Inactive   | 0.98        |
| Tox21-Stress response pathways             | <a href="#">Mitochondrial Membrane Potential (MMP)</a>                                                | sr_mmp        | Inactive   | 0.93        |
| Tox21-Stress response pathways             | <a href="#">Phosphoprotein (Tumor Suppressor) p53</a>                                                 | sr_p53        | Inactive   | 0.90        |
| Tox21-Stress response pathways             | <a href="#">ATPase family AAA domain-containing protein 5 (ATAD5)</a>                                 | sr_atad5      | Inactive   | 0.97        |
| Molecular Initiating Events                | <a href="#">Thyroid hormone receptor alpha (THRα)</a>                                                 | mie_thr_alpha | Inactive   | 0.94        |
| Molecular Initiating Events                | <a href="#">Thyroid hormone receptor beta (THRβ)</a>                                                  | mie_thr_beta  | Inactive   | 0.77        |
| Molecular Initiating Events                | <a href="#">Transthyretin (TTR)</a>                                                                   | mie_ttr       | Inactive   | 0.97        |
| Molecular Initiating Events                | <a href="#">Ryanodine receptor (RYR)</a>                                                              | mie_ryr       | Inactive   | 0.91        |

| Classification              | Target                                                               | Shorthand  | Prediction | Probability |
|-----------------------------|----------------------------------------------------------------------|------------|------------|-------------|
| Molecular Initiating Events | GABA receptor (GABAR)                                                | mie_gabar  | Active     | 0.73        |
| Molecular Initiating Events | Glutamate N-methyl-D-aspartate receptor (NMDAR)                      | mie_nmdar  | Inactive   | 0.94        |
| Molecular Initiating Events | alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionate receptor (AMPA) | mie_ampar  | Inactive   | 0.99        |
| Molecular Initiating Events | Kainate receptor (KAR)                                               | mie_kar    | Inactive   | 0.99        |
| Molecular Initiating Events | Achetylcholinesterase (AChE)                                         | mie_ache   | Active     | 0.54        |
| Molecular Initiating Events | Constitutive androstane receptor (CAR)                               | mie_car    | Inactive   | 0.98        |
| Molecular Initiating Events | Pregnane X receptor (PXR)                                            | mie_pxr    | Inactive   | 0.71        |
| Molecular Initiating Events | NADH-quinone oxidoreductase (NADHox)                                 | mie_nadhox | Inactive   | 0.98        |
| Molecular Initiating Events | Voltage-gated sodium channel (VGSC)                                  | mie_vgsc   | Inactive   | 0.74        |
| Molecular Initiating Events | Na+/I- symporter (NIS)                                               | mie_nis    | Inactive   | 0.91        |
| Metabolism                  | Cytochrome CYP1A2                                                    | CYP1A2     | Inactive   | 0.76        |
| Metabolism                  | Cytochrome CYP2C19                                                   | CYP2C19    | Inactive   | 0.77        |
| Metabolism                  | Cytochrome CYP2C9                                                    | CYP2C9     | Inactive   | 0.62        |
| Metabolism                  | Cytochrome CYP2D6                                                    | CYP2D6     | Inactive   | 0.52        |
| Metabolism                  | Cytochrome CYP3A4                                                    | CYP3A4     | Inactive   | 0.74        |
| Metabolism                  | Cytochrome CYP2E1                                                    | CYP2E1     | Inactive   | 1.0         |

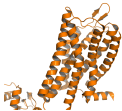
Toxicity targets

Possible binding to toxicity targets is shown below. For more information on the targets, please click on the individual abbreviations.



|       |       |      |      |       |      |      |      |     |      |      |      |      |       |      |      |
|-------|-------|------|------|-------|------|------|------|-----|------|------|------|------|-------|------|------|
| AA2AR | ADRB2 | ANDR | AOFA | CRFR1 | DRD3 | ESR1 | ESR2 | GCR | HRH1 | NR12 | OPRK | OPRM | PDE4D | PGH1 | PRGR |
|       |       |      |      |       |      |      |      |     |      |      |      |      |       |      |      |

Details about possible toxicity targets:

|                                                                                     | Toxicity Target    | Avg Pharmacophore Fit | Avg Similarity Known Ligands |
|-------------------------------------------------------------------------------------|--------------------|-----------------------|------------------------------|
|  | Opioid Receptor Mu | 0%                    | 73.42%                       |